

МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ РОССИЙСКОЙ ФЕДЕРАЦИИ

ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ БЮДЖЕТНОЕ ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
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Практическая работа на тему:
Настройка SSH и NAT

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Настроим внешний интерфейс на Router0:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Loopback0

Router(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

Router(config-if)#no shutdown'
                             ^
% Invalid input detected at '^' marker.

Router(config-if)#no shutdown
Router(config-if)#ip address 203.0.113.1 255.255.255.252
Router(config-if)#
```

SSH на Router0:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R0
R0(config)#ip domain-name company.local
R0(config)#crypto key generate rsa modulus 1024
                                   ^
% Invalid input detected at '^' marker.

R0(config)#crypto key generate rsa modulus 1024
                                   ^
% Invalid input detected at '^' marker.

R0(config)#crypto key generate rsa
The name for the keys will be: R0.company.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R0(config)#show crypto key mypubkey rsa
*Mar 1 0:30:33.83: %SSH-5-ENABLED: SSH 1.99 has been enabled
           ^
% Invalid input detected at '^' marker.

R0(config)#username admin privilege 15 secret SecurePass2025
R0(config)#line vty 0 4
R0(config-line)#login local
R0(config-line)#transport input ssh
R0(config-line)#exit
R0(config)#enable secret Enable$2025
R0(config)#service password-encryption
R0(config)#banner motd
% Incomplete command.
R0(config)#banner motd *
Enter TEXT message.  End with the character '*'.

Hello
*

R0(config)#no ip domain-lookup
R0(config)#ip ssh version 2
R0(config)#
```

Настройка Router1:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#ip domain-name company.local
R1(config)#crypto rsa
                  ^
% Invalid input detected at '^' marker.

R1(config)#crypto key generate rsa
The name for the keys will be: R1.company.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#username admin privilege 15 secret SecurePass2025
*Mar 1 0:11:59.126: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#enable secret Enable$2025
R1(config)#service password-encryption
R1(config)#banner motd *
Enter TEXT message.  End with the character '*'.
Hello
```

Настройка Router3:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#ip domain-name company.local
R3(config)#crypto key generate rsa
The name for the keys will be: R3.company.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#username admin privilege 15 secret SecurePass2025
*Mar 1 0:19:48.367: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#exit
R3(config)#enable secret Enable$2025
R3(config)#service password-encryption
R3(config)#banner motd *
Enter TEXT message.  End with the character '*'.
Hello *

R3(config)#no ip domain-lookup
R3(config)#ip ssh version 2
R3(config)#
```

Настройка Router4:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R3
R3(config)#ip domain-name company.local
R3(config)#crypto key generate rsa
The name for the keys will be: R3.company.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#username admin privilege 15 secret SecurePass2025
*Mar 1 0:19:48.367: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#exit
R3(config)#enable secret Enable$2025
R3(config)#service password-encryption
R3(config)#banner motd *
Enter TEXT message.  End with the character '*'.
Hello *

R3(config)#no ip domain-lookup
R3(config)#ip ssh version 2
R3(config)#
```

Настройка NAT на Router0:

```
R0>enable
Password:
R0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R0(config)#interface GigabitEthernet
% Incomplete command.
R0(config)#interface GigabitEthernet0/0
R0(config-if)#ip nat inside
R0(config-if)#exit
R0(config)#interface GigabitEthernet0/1
R0(config-if)#ip nat inside
R0(config-if)#exit
R0(config)#interface GigabitEthernet0/2
R0(config-if)#ip nat inside
R0(config-if)#exit
R0(config)#interface Loopback0
R0(config-if)#ip address 203.0.113.1 255.255.255.252
R0(config-if)#ip nat outside
R0(config-if)#exit
R0(config)#ip access-list standard NAT_INTERNAL
R0(config-std-nacl)#permit 192.168.1.0 0.0.0.255
R0(config-std-nacl)#permit 192.168.2.0 0.0.0.255
R0(config-std-nacl)#permit 192.168.3.0 0.0.0.255
R0(config-std-nacl)#permit 192.168.10.0 0.0.0.255
R0(config-std-nacl)#permit 192.168.20.0 0.0.0.255
R0(config-std-nacl)#permit 10.0.1.0 0.0.0.255
R0(config-std-nacl)#permit 10.0.2.0 0.0.0.255
R0(config-std-nacl)#permit 10.0.10.0 0.0.0.255
R0(config-std-nacl)#permit 172.16.1.0 0.0.0.255
R0(config-std-nacl)#permit 172.16.2.0 0.0.0.255
R0(config-std-nacl)#permit 172.16.3.0 0.0.0.255
R0(config-std-nacl)#permit 172.16.10.0 0.0.0.255
R0(config-std-nacl)#permit 172.16.20.0 0.0.0.255
R0(config-std-nacl)#exit
R0(config)#ip nat inside source list NAT_INTERNAL interface Loopback0
                                                            ^
% Invalid input detected at '^' marker.

R0(config)#ip nat pool NATPOOL 203.0.113.1 203.0.113.1 netmask 255.255.255.252
R0(config)#ip nat inside source list NAT_INTERNAL pool NATPOOL overload
R0(config)#
```

Проверка:

```
R0#show ip nat translations
R0#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.1:10    192.168.1.10:10    8.8.8.8:10         8.8.8.8:10
icmp 203.0.113.1:11    192.168.1.10:11    8.8.8.8:11         8.8.8.8:11
icmp 203.0.113.1:12    192.168.1.10:12    8.8.8.8:12         8.8.8.8:12
icmp 203.0.113.1:8     192.168.1.10:8     8.8.8.8:8          8.8.8.8:8
icmp 203.0.113.1:9     192.168.1.10:9     8.8.8.8:9          8.8.8.8:9
```